VISUAL JIGSAW POST-TRAINING IMPROVES MLLMS

Anonymous Author

ABSTRACT

Reinforcement learning (RL)-based post-training has become the dominant paradigm for improving the reasoning capabilities of large language models, yet its application to multimodal large language models (MLLMs) has remained largely text-centric, focusing on mathematical and logical reasoning tasks that treat visual inputs merely as context for textual inference. We argue that visual perception itself—the ability to understand spatial layouts, temporal dynamics, and three-dimensional structure—deserves direct optimization. To this end, we introduce **Visual Jigsaw (VJ)**, a simple yet effective post-training framework that recasts visual understanding as a verifiable ordering task: given shuffled visual elements, the model must predict the correct permutation. We instantiate this idea across three modalities: (1) **Image Jigsaw**, which shuffles patches of a 3×3 grid; (2) **Video Jigsaw**, which shuffles temporal clips; and (3) **3D Jigsaw**, which shuffles depth-ordered RGB-D point sets. Trained with Group Relative Policy Optimization (GRPO) and a graded reward function that provides partial credit for approximately correct orderings, Visual Jigsaw post-training yields consistent improvements across a wide range of benchmarks. Applied to Qwen2.5-VL-7B, our approach achieves gains of +6.00 on MMVP, +6.06 on MMStar fine-grained, +5.23 on AoTBench, and +17.11 on DA-2K for 3D understanding, while operating as a lightweight, plug-and-play module that preserves existing reasoning capabilities.

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1 INTRODUCTION

The recent success of reinforcement learning from verifiable rewards (RLVR) in improving the reasoning capabilities of large language models Guo et al. (2025); Shao et al. (2024) has prompted a wave of efforts to extend similar techniques to multimodal large language models (MLLMs) Huang et al. (2025); Liu et al. (2025b); Meng et al. (2025); Wang et al. (2025e); Yuan et al. (2025a). However, a critical limitation persists: nearly all existing approaches optimize for text-centric reasoning tasks such as mathematical problem solving Lu et al. (2023); Zhang et al. (2024b) and logical inference, using visual inputs only as passive context for generating textual answers. The visual perception capabilities of MLLMs—understanding spatial relationships, temporal ordering, and three-dimensional structure—remain largely unoptimized by post-training.

This gap is consequential. Despite rapid progress in MLLM architectures Bai et al. (2025); Chen et al. (2024b); Li et al. (2024), recent evaluations reveal persistent weaknesses in fine-grained visual understanding. Models struggle with tasks requiring precise spatial reasoning Tong et al. (2024); Liu et al. (2023), temporal comprehension in videos Xue et al. (2025); Zhang et al. (2024a), and 3D spatial awareness Ray et al. (2024); Ma et al. (2024). These deficiencies suggest that scaling model size and pre-training data alone is insufficient; targeted post-training that directly exercises visual perception is needed.

We draw inspiration from a classic self-supervised learning paradigm: the jigsaw puzzle Noroozi & Favaro (2016). Solving a jigsaw puzzle requires understanding spatial relationships between parts of an image—precisely the kind of visual reasoning we wish to strengthen. However, unlike prior work that used jigsaw tasks to learn visual representations in encoders Noroozi & Favaro (2016); Carlucci et al. (2019); Wei et al. (2019), we reformulate jigsaw solving as a *language generation* task suitable for RL-based post-training of MLLMs. Given shuffled visual elements, the model must output the correct permutation as a sequence of indices. This formulation has two key advantages: (1) the answer is easily verifiable, enabling RLVR training without human annotation, and (2)

the task directly exercises visual perception through the model's existing autoregressive generation mechanism, requiring no additional modules.

We introduce **Visual Jigsaw (VJ)**, a post-training framework that instantiates this idea across three visual modalities (Figure 2):

- **Image Jigsaw**: An image is divided into a 3×3 grid and the patches are shuffled. The model must predict the original ordering from the shuffled patches, using images from COCO Lin et al. (2014).
- **Video Jigsaw**: A video is split into $K{=}4$ temporal clips that are shuffled. The model must recover the chronological order, using videos from LLaVA-Video Zhang et al. (2024c).
- **3D Jigsaw**: RGB-D point sets from ScanNet Dai et al. (2017) are divided into depth-ordered regions and shuffled. The model must predict the correct depth ordering.

Unifying these three modalities under a single framework is essential because spatial, temporal, and depth reasoning represent complementary axes of visual understanding that are rarely exercised together. Moreover, RL-based training is well-suited here: the combinatorial nature of permutation prediction makes supervised fine-tuning prone to memorization, whereas RL with partial-credit rewards encourages genuine perceptual reasoning by rewarding incremental ordering improvements.

Training is performed using Group Relative Policy Optimization (GRPO) Shao et al. (2024) with a graded reward function that provides exact-match rewards, partial credit for approximately correct orderings, and penalties for degenerate outputs. This reward shaping is critical: it provides a smoother learning signal than binary rewards, enabling the model to make incremental progress on the combinatorially challenging ordering task.

Our experiments demonstrate that Visual Jigsaw post-training yields broad and consistent improvements. Applied to Qwen2.5-VL-7B Bai et al. (2025), Image Jigsaw improves performance on MMVP Tong et al. (2024) by +6.00 and MMStar fine-grained Chen et al. (2024a) by +6.06. Video Jigsaw improves AoTBench Xue et al. (2025) by +5.23 and Vinoground Zhang et al. (2024a) by +3.40. 3D Jigsaw achieves remarkable gains of +17.11 on DA-2K Yang et al. (2024) and +15.34 on SAT Ray et al. (2024). Notably, RL-based training consistently outperforms supervised fine-tuning (SFT) on the same jigsaw data, and Visual Jigsaw is complementary to existing reasoning-focused post-training: when applied to ThinkLite-VL Wang et al. (2025e), it improves vision-centric performance while preserving mathematical reasoning capabilities.

2 Related Works

Multimodal Large Language Models. The rapid development of MLLMs has produced a series of increasingly capable architectures. Early models such as LLaVA Liu et al. (2024b) demonstrated the viability of combining pre-trained vision encoders with large language models through visual instruction tuning. Subsequent work has scaled both model capacity and training data, leading to models such as InternVL Chen et al. (2024c;b); Zhu et al. (2025a), LLaVA-NeXT Liu et al. (2024a), LLaVA-OneVision Li et al. (2024), and the Qwen2-VL family Wang et al. (2024); Bai et al. (2025). These models have achieved strong performance across a range of multimodal benchmarks, with the most recent iterations supporting multi-image, video, and even 3D inputs. Commercial models such as GPT-4o Hurst et al. (2024) and Gemini 2.5 Comanici et al. (2025) have further pushed the frontier. Despite these advances, systematic evaluations continue to reveal weaknesses in fine-grained visual understanding Tong et al. (2024); Chen et al. (2024a), suggesting that architectural and data scaling alone are insufficient.

RL-Based Post-Training for MLLMs. Reinforcement learning from human feedback (RLHF) Ouyang et al. (2022) and its variants such as Direct Preference Optimization (DPO) Rafailov et al. (2023) were among the first post-training methods applied to language models. The success of DeepSeek-R1 Guo et al. (2025) in improving reasoning through RL with verifiable rewards (RLVR) using Group Relative Policy Optimization (GRPO) Shao et al. (2024) has sparked significant interest in applying similar techniques to MLLMs. Vision-R1 Huang et al. (2025), Visual-RFT Liu et al. (2025b), and MM-Eureka Meng et al. (2025) apply RLVR to multimodal mathematical and scientific reasoning. ThinkLite-VL Wang et al. (2025e) and VL-Cogito Yuan et al. (2025a) focus on improv-

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ing chain-of-thought reasoning in MLLMs. More domain-specific efforts include Seg-Zero Liu et al. (2025a) for segmentation, Video-R1 Feng et al. (2025) for video understanding, Scene-R1 Yuan et al. (2025b) and DeepEyes Zheng et al. (2025) for spatial reasoning, and ASVR Wang et al. (2025a) and ROSS Wang et al. (2025b) for spatial and 3D understanding. However, the vast majority of these approaches optimize for text-centric reasoning tasks, using visual inputs as context rather than directly training visual perception. Our work differs fundamentally by designing a visual task—jigsaw puzzle solving—that directly exercises spatial, temporal, and 3D understanding.

Jigsaw Puzzles as Self-Supervised Learning. The use of jigsaw puzzles as a pretext task for self-supervised visual representation learning was pioneered by Noroozi and Favaro Noroozi & Favaro (2016), who trained convolutional networks to predict the permutation of shuffled image patches. This idea was extended to domain generalization Carlucci et al. (2019), iterative puzzle solving Wei et al. (2019), fine-grained recognition Du et al. (2020), and medical imaging Chen et al. (2023). In the video domain, temporal ordering tasks have been used as self-supervised signals for learning video representations Ahsan et al. (2019); Huo et al. (2021); Wang et al. (2022). More broadly, self-supervised methods based on predicting spatial or structural properties of images—including rotation prediction Gidaris et al. (2018), relative patch position Doersch et al. (2015), masked image modeling He et al. (2022a); Bao et al. (2021); He et al. (2022b); Zhou et al. (2021), and contrastive learning He et al. (2020); Caron et al. (2021); Oquab et al. (2023); Assran et al. (2023)—have been foundational to modern visual representation learning. Our work revisits the jigsaw paradigm in a fundamentally different context: rather than training a visual encoder, we use jigsaw solving as an RL reward signal for post-training an MLLM, formulating the task as text generation. The concurrent work of Jigsaw-R1 Wang et al. (2025f) shares the idea of using jigsaw puzzles for MLLM post-training but focuses exclusively on images; our framework unifies image, video, and 3D modalities and introduces a graded reward function that provides partial credit.

3 METHOD

We present Visual Jigsaw (VJ), a post-training framework that improves the visual perception capabilities of MLLMs through reinforcement learning on jigsaw puzzle tasks. The key idea is to shuffle visual elements—image patches, video clips, or 3D point sets—and train the model to predict the correct ordering using GRPO with a graded reward function. Figure 2 provides an overview of the framework.

3.1 VISUAL JIGSAW TASK FORMULATION

Given a visual input $\mathcal{V}$, we decompose it into N ordered elements $\{v_1, v_2, \ldots, v_N\}$ according to a modality-specific decomposition strategy. We then apply a random permutation σ to obtain the shuffled sequence $\{v_{\sigma(1)}, v_{\sigma(2)}, \ldots, v_{\sigma(N)}\}$, which is presented to the model along with a text prompt instructing it to recover the original ordering. The model must output the permutation σ as a sequence of indices enclosed in a `\boxed{}` environment, preceded by a chain-of-thought reasoning trace enclosed in `<think>...</think>` tags.

We instantiate this formulation across three visual modalities:

Image Jigsaw. Given an image I, we divide it into a 3×3 grid of non-overlapping patches, yielding $N{=}9$ patches. The patches are randomly shuffled and presented to the model as a composite image with visible grid lines. The model must predict the permutation that maps each shuffled patch to its original position. We use images from COCO Lin et al. (2014) as the source data.

Video Jigsaw. Given a video V, we uniformly sample $K{=}4$ non-overlapping temporal clips. The clips are shuffled and presented to the model as a sequence of video segments. The model must predict the chronological ordering of the clips. We use videos from LLaVA-Video Zhang et al. (2024c) as the source data.

3D Jigsaw. Given an RGB-D scan from ScanNet Dai et al. (2017), we partition the point cloud into regions ordered by depth. The regions are shuffled and presented to the model as a sequence of RGB-D images. The model must predict the correct depth ordering.

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3.2 TRAINING WITH GRPO

We train the model using Group Relative Policy Optimization (GRPO) Shao et al. (2024), which estimates advantages from a group of sampled responses without requiring a separate value network. For each jigsaw prompt x, we sample a group of G responses $\{y_1, y_2, \ldots, y_G\}$ from the current policy π_θ. The GRPO objective is:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{x\sim\mathcal{D}}\left[\frac{1}{G}\sum_{i=1}^{G}\min\left(\frac{\pi_\theta(y_i|x)}{\pi_{\theta_{\text{old}}}(y_i|x)}\hat{A}_i,\ \text{clip}\left(\frac{\pi_\theta(y_i|x)}{\pi_{\theta_{\text{old}}}(y_i|x)}, 1-\epsilon, 1+\epsilon\right)\hat{A}_i\right)\right], \quad (1)$$

where $\hat{A}_i$ is the advantage estimated by normalizing the rewards within the group:

$$\hat{A}_i = \frac{r_i - \text{mean}(\{r_j\}_{j=1}^{G})}{\text{std}(\{r_j\}_{j=1}^{G})}. \quad (2)$$

We use the verl framework Sheng et al. (2024) for training with Qwen2.5-VL-7B-Instruct Bai et al. (2025) as the backbone. Training is performed on 8 GPUs with a learning rate of 1×10^{-6}, a rollout group size of G=16, a batch size of 256 for Image Jigsaw (128 for Video and 3D Jigsaw), and 32 training epochs.

3.3 GRADED REWARD FUNCTION

A critical design choice is the reward function. Binary reward—1 for a correct permutation and 0 otherwise—provides an extremely sparse learning signal, since the probability of randomly guessing a correct permutation of N elements is $1/N!$. For N=9 (Image Jigsaw), this is approximately 2.8×10^{-6}. We therefore design a graded reward function r that provides partial credit:

$$r(y) = r_{\text{format}}(y) + r_{\text{acc}}(y) + r_{\text{rep}}(y), \quad (3)$$

where each component is defined as follows:

Format reward r_{format}. We assign $r_{\text{format}} = 0.2$ if the response follows the required format with `<think>...</think>` reasoning and a `\boxed{}` answer, and $r_{\text{format}} = 0$ otherwise.

Accuracy reward r_{acc}. If the predicted permutation exactly matches the ground truth, we assign $r_{\text{acc}} = 1.0$. If the predicted permutation has the correct length, contains unique indices, but is not fully correct, we provide partial credit proportional to the number of correctly placed elements: $r_{\text{acc}} = 0.2 \times (\text{number of correct positions}/N)$, applied only when $N > 4$ to avoid rewarding near-random guesses on short sequences.

Repetition penalty r_{rep}. To discourage degenerate outputs with excessive token repetition, we apply a penalty of $r_{\text{rep}} = -0.5$ when repetitive patterns are detected in the response.

This graded reward provides a smoother optimization landscape than binary rewards, enabling the model to receive learning signal from partially correct solutions and gradually improve toward exact matches.

4 EXPERIMENTS

We evaluate Visual Jigsaw post-training across image, video, and 3D understanding benchmarks. Our backbone model is Qwen2.5-VL-7B-Instruct Bai et al. (2025), and we compare against the unmodified baseline as well as recent RL-based post-training methods.

4.1 IMAGE UNDERSTANDING

Setup. We train Image Jigsaw using 3×3 grid patches from COCO images Lin et al. (2014) and evaluate on 13 image understanding benchmarks spanning fine-grained perception Tong et al. (2024); Chen et al. (2024a); Wang et al. (2025c), spatial reasoning Liu et al. (2023); Yao et al. (2024); Jia et al. (2025), grounding Lai et al. (2024), and compositional understanding Thrush et al. (2022); Dumpala et al. (2024).

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Results. As shown in Table 1, Image Jigsaw (RL) achieves consistent improvements over the Qwen2.5-VL-7B baseline across nearly all benchmarks. The most substantial gains are observed on fine-grained visual perception tasks: +6.00 on MMVP Tong et al. (2024), +6.06 on MMStar fine-grained Chen et al. (2024a), and +5.90 on DA-2K Yang et al. (2024). Spatial reasoning benchmarks also benefit, with improvements of +2.68 on VSR Liu et al. (2023) and +1.83 on OmniSpatial Jia et al. (2025). On general benchmarks such as MMBench Liu et al. (2024c), the improvement is modest (+1.12), suggesting that Visual Jigsaw specifically targets fine-grained perception rather than broad knowledge. Notably, Image Jigsaw also outperforms recent RL post-training approaches including ThinkLite-VL Wang et al. (2025e), VL-Cogito Yuan et al. (2025a), and LLaVA-Critic-R1 Wang et al. (2025d) on vision-centric benchmarks, despite these methods being trained on substantially more diverse task data.

Table 1: **Evaluation results on image benchmarks.** Image Jigsaw achieves consistent improvements across fine-grained perception, spatial understanding, and compositional understanding tasks.

	Fine-grained Perception & Understanding								Spatial Und (Mono)			Compositional Und	
Model	**MMVP**	**MMStar (fine-grained)**	**MMBench**	**HR-Bench-8K**	**V***	**MME-RealWorld**	**LISA-Grounding**	**OVD-Eval**	**VSR**	**OmniSpatial**	**DA-2K**	**Winoground**	**SugarCrepe++**
	test	fine	en_dev	test	test	lite	test	test	test	test	val	g-acc	test
ThinkLite-VL	55.33	59.95	84.19	68.12	76.96	46.17	73.70	35.78	78.09	42.60	58.46	35.25	61.49
VL-Cogito	55.33	56.64	82.98	69.62	79.58	47.63	72.26	35.78	79.82	44.29	56.43	38.25	63.59
LLaVA-Critic-R1	53.33	57.80	83.16	67.50	78.01	45.18	68.52	35.28	78.50	42.73	53.82	34.75	61.93
Qwen2.5-VL-7B	54.66	59.75	83.33	67.38	76.96	43.41	71.89	35.07	77.68	42.66	54.45	37.00	61.59
Image Jigsaw (SFT)	56.00	60.94	83.67	69.75	80.10	43.88	66.59	34.35	80.68	43.55	61.46	38.75	62.03
Image Jigsaw	60.66	65.81	84.45	71.13	80.63	45.96	74.54	36.49	80.36	44.49	60.35	39.00	63.02
(Gain)	+6.00	+6.06	+1.12	+3.75	+3.66	+2.55	+2.65	+1.42	+2.68	+1.83	+5.90	+2.00	+1.43

RL vs. SFT. A key finding is that RL-based training consistently outperforms SFT on the same jigsaw data. While SFT also yields some improvements on perception benchmarks, it degrades performance on grounding tasks—LISA-Grounding drops from 71.89 to 66.59 under SFT, while RL improves it to 74.54. This result is consistent with prior observations that SFT can lead to catastrophic forgetting of existing capabilities Chen et al. (2025); Huan et al. (2025), whereas RL-based training better preserves the model's original abilities while acquiring new ones.

4.2 Video Understanding

Setup. We train Video Jigsaw using $K{=}4$ shuffled temporal clips from LLaVA-Video Zhang et al. (2024c) and evaluate on 13 video understanding benchmarks with 64-frame inputs.

Results. Table 2 presents the results. Video Jigsaw yields the largest improvements on temporal reasoning benchmarks: +5.23 on AoTBench Xue et al. (2025), +3.40 on Vinoground Zhang et al. (2024a), and +3.00 on CVBench Zhu et al. (2025b). These are precisely the benchmarks that require understanding temporal order and dynamics, validating that the temporal ordering task in Video Jigsaw directly strengthens temporal perception. Improvements are also observed across general video understanding benchmarks, including +1.30 on VideoMME Fu et al. (2025), +1.62 on MotionBench Hong et al. (2025), and +2.66 on VSI-Bench Yang et al. (2025). Even benchmarks less directly related to temporal ordering show positive gains, indicating that improved temporal understanding transfers to broader video comprehension.

4.3 3D Understanding

Setup. We train 3D Jigsaw using depth-ordered RGB-D point sets from ScanNet Dai et al. (2017) and evaluate on 8 benchmarks that assess 3D spatial reasoning.

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Table 2: **Evaluation results on video benchmarks.** Video Jigsaw consistently improves over the baseline across all benchmarks and frame settings.

Model	Frames	AoTBench	Vinoground	TOMATO	FAVOR-Bench	TUNA-Bench	VideoMME	TempCompass	TVBench	MotionBench	LVBench	VSI-Bench	Video-TT	CVBench
		vqa	group	test	test	test	wo subs	mc	test	val	test	test	mcq	test
Video-R1	16	45.06	9.40	27.29	49.47	53.00	56.62	70.19	51.80	55.82	34.53	34.34	42.95	47.50
Video-R1	32	47.53	10.20	27.29	49.90	54.26	59.88	71.77	53.54	56.12	38.61	35.11	42.63	48.10
Video-R1	64	48.68	10.60	27.36	50.51	54.33	60.85	72.59	53.43	56.09	38.80	36.61	42.74	48.69
Qwen2.5-VL-7B	16	45.52	12.60	25.87	48.54	53.14	57.44	71.77	49.94	55.56	33.51	32.79	38.39	47.70
Qwen2.5-VL-7B	32	49.48	18.20	26.34	49.34	54.88	60.70	72.59	51.96	56.47	39.19	35.34	41.57	49.60
Qwen2.5-VL-7B	64	52.41	21.80	26.35	50.86	55.79	63.44	72.84	53.74	56.29	40.35	37.74	42.25	51.50
Video Jigsaw	16	51.67	15.20	27.56	49.69	55.10	58.07	73.10	51.33	56.87	36.41	35.39	40.19	49.80
(Gain)		+6.15	+2.60	+1.69	+1.15	+1.96	+0.63	+1.33	+1.39	+1.31	+2.90	+2.60	+1.80	+2.10
Video Jigsaw	32	55.00	21.40	28.03	50.56	56.49	62.37	73.60	53.31	57.99	39.70	38.47	43.27	51.60
(Gain)		+5.52	+3.20	+1.69	+1.22	+1.61	+1.67	+1.01	+1.35	+1.52	+0.51	+3.13	+1.70	+2.00
Video Jigsaw	64	57.64	25.20	28.30	52.27	56.63	64.74	73.60	54.18	57.91	41.83	40.40	44.11	54.50
(Gain)		+5.23	+3.40	+1.95	+1.41	+0.84	+1.30	+0.76	+0.44	+1.62	+1.48	+2.66	+1.86	+3.00

Results. As shown in Table 3, 3D Jigsaw achieves the most dramatic improvements among all three modalities. DA-2K Yang et al. (2024) improves by +17.11 (from 54.45 to 71.56), and SAT Ray et al. (2024) improves by +15.34 (from 48.66 to 64.00). These large gains suggest that the baseline model has substantial untapped potential for 3D understanding that depth-ordering training effectively unlocks. Consistent improvements are also observed on OmniSpatial Jia et al. (2025) (+3.33), VSI-Bench Yang et al. (2025) (+2.90), SPARBench Zhang et al. (2025) (+2.56), ViewSpatial Li et al. (2025) (+2.10), and All-Angles Yeh et al. (2025) (+1.50).

Table 3: **Evaluation results on 3D benchmarks.** 3D Jigsaw consistently enhances performance on both directly related depth comparison tasks (DA-2K) and broader 3D perception tasks spanning single-view, multi-view, and egocentric video inputs.

Model	SAT-Real	3DSRBench	ViewSpatial	All-Angles	OmniSpatial	VSI-Bench	SPARBench	DA-2K
	test	test	test	test	test	test	tiny	test
Qwen2.5-VL-7B	48.66	57.42	36.52	47.56	42.66	37.74	35.75	54.45
3D Jigsaw	64.00	58.13	38.62	49.06	45.99	40.64	38.31	71.56
(Gain)	+15.34	+0.71	+2.10	+1.50	+3.33	+2.90	+2.56	+17.11

4.4 Complementarity with Reasoning Post-Training

An important question is whether Visual Jigsaw post-training is compatible with existing reasoning-focused post-training. To investigate this, we apply Image Jigsaw to ThinkLite-VL Wang et al. (2025e), a model already post-trained for enhanced chain-of-thought reasoning. As shown in Table 4, Image Jigsaw improves the average vision-centric score by +1.91 while largely preserving mathematical reasoning performance: MathVista Lu et al. (2023) remains stable at 75.10 (vs. 75.20), MathVerse Zhang et al. (2024b) at 50.50 (vs. 50.76), and MMMU Yue et al. (2024) shows only a slight decrease of -0.89. Meanwhile, MathVision improves by +4.28 and EMMA Hao et al. (2025) by +2.25, suggesting that improved visual perception can even benefit some reasoning tasks. This result confirms that Visual Jigsaw is complementary to reasoning-focused post-training and can be applied as a plug-and-play enhancement.

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Table 4: Performance of Visual Jigsaw on a reasoning-oriented MLLM (ThinkLite-VL), showing improved visual perception while preserving reasoning ability.

Model	Vision-Centric Avg	MathVista testmini	MathVision testmini	MathVerse testmini	MMMU val	EMMA mini
ThinkLite-VL	59.69	75.20	30.92	50.76	55.11	26.75
+ Image Jigsaw	61.60	75.10	35.20	50.50	54.22	29.00

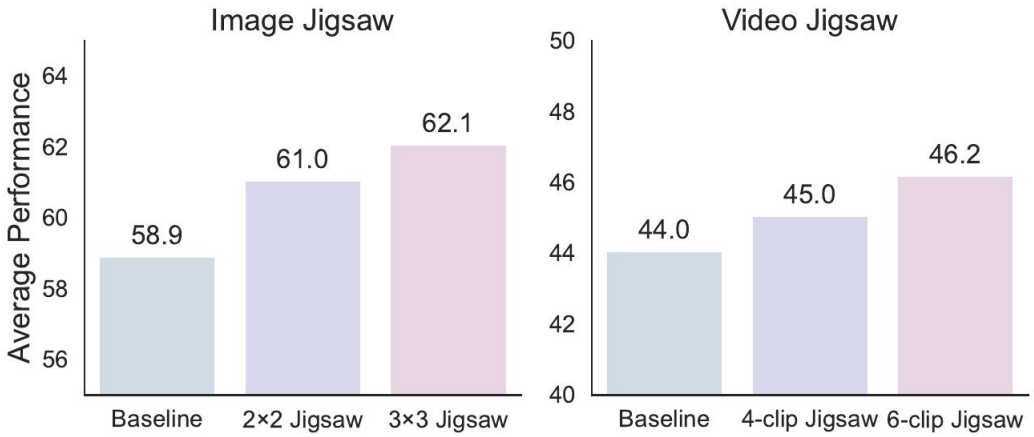

Figure 1: Effect of jigsaw difficulty on downstream performance. *(Left)* Image Jigsaw with varying grid sizes. *(Right)* Video Jigsaw with varying numbers of temporal clips.

4.5 Ablation Study

Effect of jigsaw difficulty. We study the effect of grid size for Image Jigsaw and number of clips for Video Jigsaw on downstream performance (Figure 1). For images, increasing from 2×2 to 3×3 grids yields consistent improvements, as finer granularity requires more precise spatial understanding. For videos, we find that K=4 clips provides a good balance between task difficulty and learnability; fewer clips make the task trivially easy for the model, while more clips introduce excessive ambiguity as visually similar segments become nearly indistinguishable. We also find that the graded reward function is critical: replacing it with binary (correct/incorrect) rewards leads to reward sparsity and slower convergence, as most sampled permutations receive zero reward early in training.

Effect of reward function design. We ablate the components of the graded reward function by training Image Jigsaw variants with each component removed. Removing the format reward r_{format} leads to a moderate decrease in benchmark scores, as the model occasionally generates malformed outputs. More critically, removing the partial-credit accuracy reward and using only binary rewards causes a significant drop in training efficiency: the model converges significantly slower and achieves lower final performance, confirming the importance of smooth reward shaping for combinatorial tasks. The repetition penalty r_{rep} proves particularly important in early training stages, preventing the model from collapsing to degenerate repetitive outputs.

Effect of training data scale. We investigate how training data volume affects Visual Jigsaw by varying the number of COCO images used for Image Jigsaw. Performance improves steadily as the training set grows from 5K to 50K images, after which gains plateau, indicating a saturation point beyond which additional data provides diminishing returns. Notably, the model exhibits strong data efficiency: even with only 10K training samples, we observe meaningful improvements over the baseline across vision-centric benchmarks. A similar trend holds for Video Jigsaw and 3D Jigsaw, where limited training data still yields substantial gains. This suggests that the jigsaw task provides a high-quality learning signal for spatial and temporal reasoning, making Visual Jigsaw practical to deploy even when large-scale training data is unavailable.

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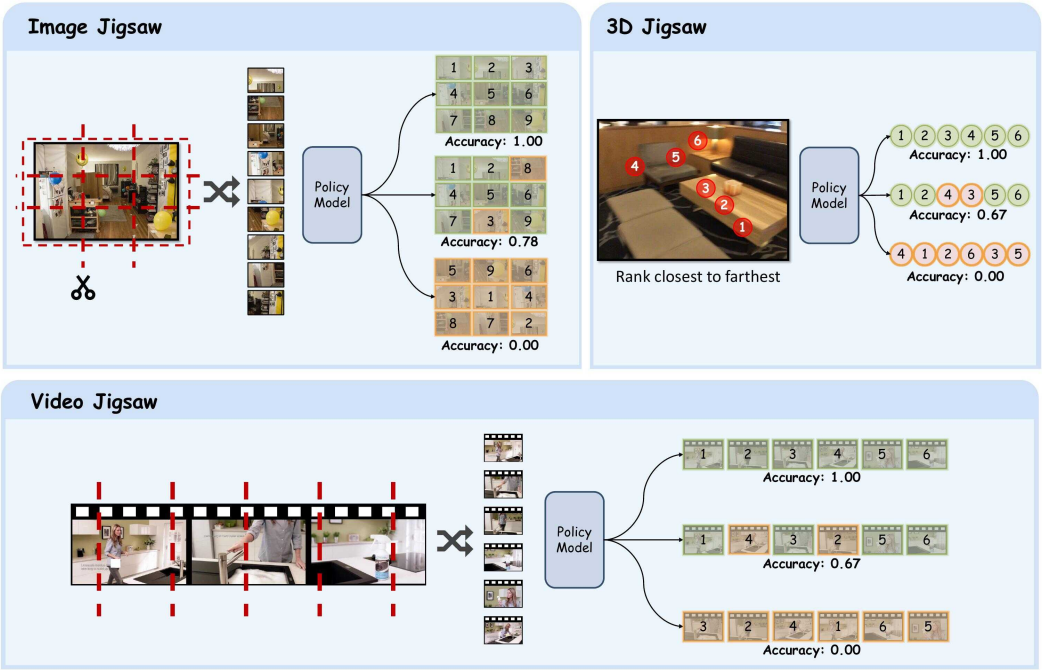

Figure 2: Overview of the Visual Jigsaw framework. We instantiate jigsaw puzzle tasks across three modalities: *(a)* Image Jigsaw shuffles patches from a 3×3 grid, *(b)* Video Jigsaw shuffles temporal clips, and *(c)* 3D Jigsaw shuffles depth-ordered RGB-D point sets. The model is trained with GRPO to predict the correct ordering.

5 Conclusion

We have presented Visual Jigsaw, a post-training framework that improves the visual perception capabilities of MLLMs by training them to solve ordering tasks on shuffled visual elements. By instantiating jigsaw puzzles across image patches, video clips, and 3D point sets, and training with GRPO using a graded reward function, our approach yields consistent improvements on a wide range of benchmarks while preserving existing capabilities. Our results demonstrate that RL-based post-training for visual perception is a promising and underexplored direction, complementary to the text-centric reasoning improvements pursued by most prior work. One limitation is that each modality is currently trained independently; joint multi-modal jigsaw training may yield further synergies, as shared spatial reasoning curricula could enable cross-modal transfer between image, video, and 3D domains. Future work could also extend the framework to dense prediction tasks such as detection and segmentation, as well as to additional modalities including audio spectrograms and multi-view image sets.

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